DATA SCIENCE & ANALYTICS

MODULE 03 SESSION 12

Predicting a category, and judging the prediction

Which customers will leave, and who should get an offer?

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Mon 19 Oct 2026

BY THE END OF TODAY, YOU CAN

Class objectives

01 Train LOGISTIC_REG and read a probability

02 Turn a probability into a decision

03 Read a confusion matrix in business words

04 Judge a model against the majority guess

05 Let the costs choose the threshold

EVERYONE DO

Step 1.1

Warm-up

Five questions on D11.

Suggested time · 10 minutes

PART 01

A first classifier, before the theory

7,043 telecom customers: upload, look, train on one feature, ask it about five customers.

INSTRUCTOR DO

Step 2.1

Today’s question

A yes-or-no question, and the table we answer it from.

Suggested time · 4 minutes

Which customers will leave this month, and who should get a retention offer?

FIVE OF 7,043 CUSTOMERS · TELCO CHURN

The column we want is Yes or No

customerID	tenure	Contract	MonthlyCharges	Churn
7590-VHVEG	1	Month-to-month	29.85	**No**
5575-GNVDE	34	One year	56.95	**No**
3668-QPYBK	2	Month-to-month	53.85	**Yes**
7795-CFOCW	45	One year	42.30	**No**
9237-HQITU	2	Month-to-month	70.70	**Yes**

Churn = Yes: the customer left within the last month. Yesterday the label was a number; today a category.

EVERYONE DO

Step 2.2

Upload and look

Check the types, count the leavers, compare contracts.

Suggested time · 14 minutes

labs/D12/01-Evr-First-Classifier/README.md

One number column arrived as text

SAFE_CAST finds the values that are not numbers.

```
SELECT tenure, TotalCharges
FROM ml.telco
WHERE SAFE_CAST(TotalCharges AS FLOAT64) IS NULL;
```

```
→ 11 rows · tenure 0 · TotalCharges blank
```

We leave TotalCharges out of today’s models: it is tenure × MonthlyCharges, roughly.

Who leaves?

COUNTIF counts the rows where the condition is true.

```
SELECT Contract,
       COUNT(*) AS customers,
       ROUND(COUNTIF(Churn = 'Yes') / COUNT(*), 3) AS share_left
FROM ml.telco
GROUP BY Contract;
```

→ 3 rows · 0.427 · 0.113 · 0.028

Month-to-month, one year, two year: the longer the contract, the fewer leave.

ALL 7,043 CUSTOMERS

1,869 left; 5,174 stayed

LEFT

26.5%

1,869 customers

STAYED

73.5%

5,174 customers

Month-to-month customers left at 43%; two-year contracts at 3%.

EVERYONE DO

Step 2.3

Train on tenure, and ask it about five customers

One feature, a probability out.

Suggested time · 14 minutes

labs/D12/01-Evr-First-Classifier/README.md

BLOCK 3 · TRAIN

One feature, a yes-or-no label

`model_type = 'LOGISTIC_REG'`: predicts a category, with a probability.

```
CREATE OR REPLACE MODEL ml.churn_tenure
OPTIONS (
  model_type = 'LOGISTIC_REG',
  input_label_cols = ['Churn'],
  data_split_method = 'RANDOM',
  data_split_eval_fraction = 0.2
) AS
SELECT tenure, Churn
FROM ml.telco;
```

Five customers, five probabilities

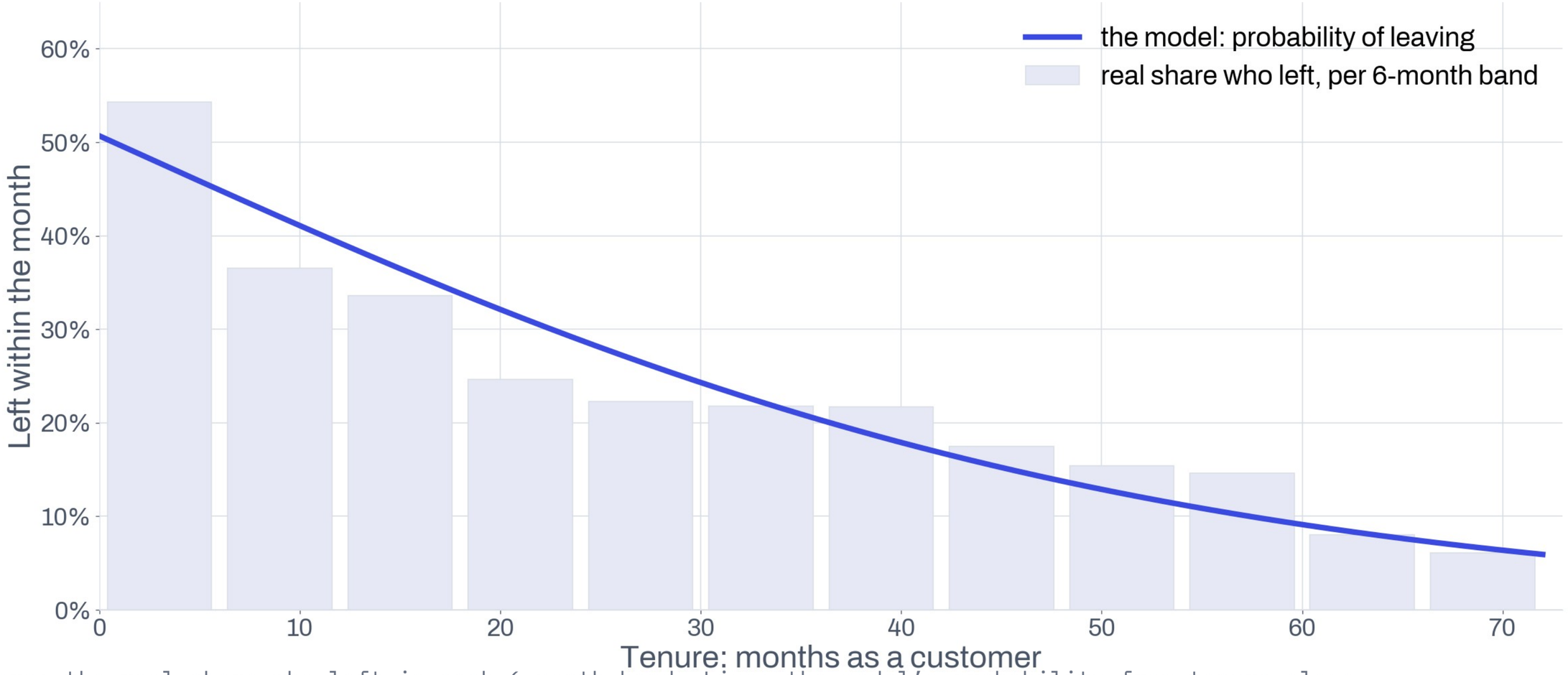
UNNEST: one row per class; we keep the probability of Yes.

```
SELECT customerID, tenure, Churn,
       ROUND(p.prob, 3) AS prob_leave
FROM ML.PREDICT(MODEL ml.churn_tenure,
       (SELECT * FROM ml.telco LIMIT 5)),
     UNNEST(predicted_Churn_probs) AS p
WHERE p.label = 'Yes';
```

→ tenure 1 ≈ 0.50 · 34 ≈ 0.22 · 45 ≈ 0.15

THE ONE-FEATURE MODEL

The longer they stay, the less likely they leave



Bars: the real share who left in each 6-month band. Line: the model's probability from tenure alone.

THE QUESTION FOR THE NEXT HOUR

A probability is not a decision. Who gets the offer at 0.49? At 0.22? Someone has to choose.

PART 02

What did it just do?

A category and a probability, the threshold, four counts, precision and recall.

INSTRUCTOR DO

Step 3.1

A category, not a number

Why not yesterday’s straight line, and the words for today.

Suggested time · 6 minutes

QUESTION



Churn is 0 or 1. Why not fit a linear regression to it?

A straight line can predict below 0 or above 1. Those values cannot be probabilities. Logistic regression turns its score into a number between 0 and 1.

TWO KINDS OF SUPERVISED LEARNING

Regression or classification?

Ask what the label holds: a number, or a group?

REGRESSION

Label: a number

Predicts how much

House price

CLASSIFICATION

Label: a category

Predicts which group

Churn yes or no

YESTERDAY'S WORDS, TODAY'S TABLE

The same words, and one new one

Word	Today
Example	one customer
Feature	tenure; later contract, charges, internet, payment …
Label	Churn: Yes or No
Positive class	**Yes, left: the outcome we act on**
Model	ml.churn_tenure: a set of fitted numbers
Prediction	a probability, and a Yes or No

INSTRUCTOR DO

Step 3.2

From a score to a probability

A weighted sum, squeezed between 0 and 1; then training and inference.

Suggested time · 8 minutes

DEFINITION

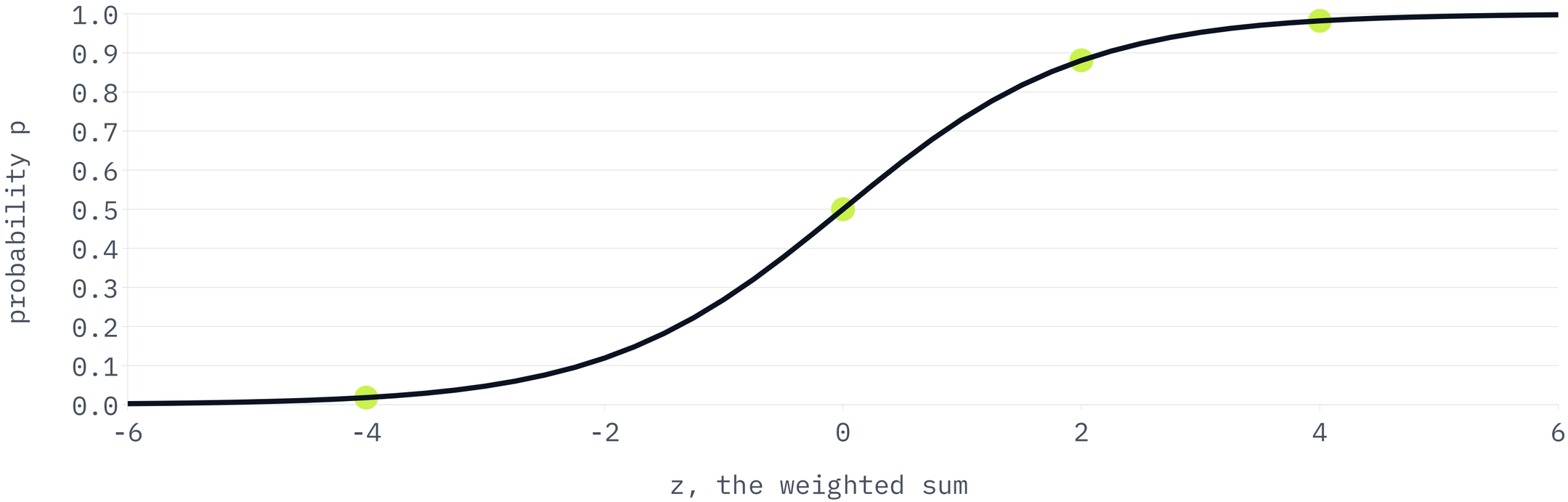
Logistic regression computes a weighted sum z, like linear regression, then turns z into a probability between 0 and 1.

The function that does the squeezing is the sigmoid. A classifier, despite the name; ours is binary.

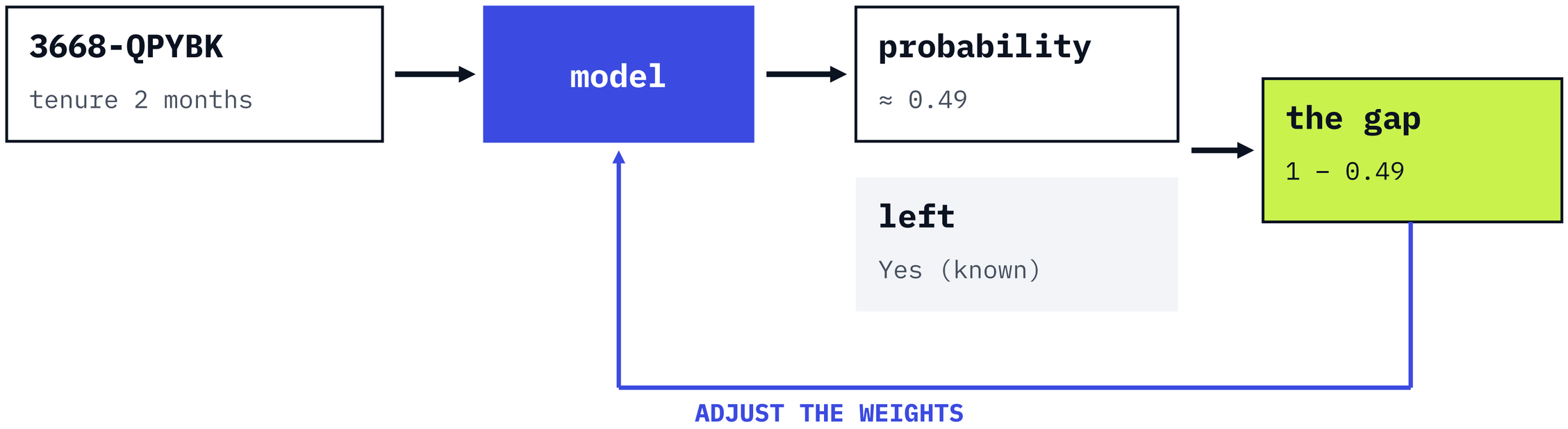
THE SIGMOID

p = 1 / (1 + e^−z)

z can be any number; the sigmoid maps it into 0 to 1. Lit: z −4, 0, 2, 4 give 0.018, 0.5, 0.881, 0.982.



Training: compare with the known answer



Training keeps the weights that make the known answers of the training customers most likely.

INFERENCE · A NEW CUSTOMER

Inference: features in, a probability out



Only the features go in. A probability is the model's estimate, not a promise.

INSTRUCTOR DO

Step 3.3

The threshold turns a probability into a decision

The same probabilities, cut in two places.

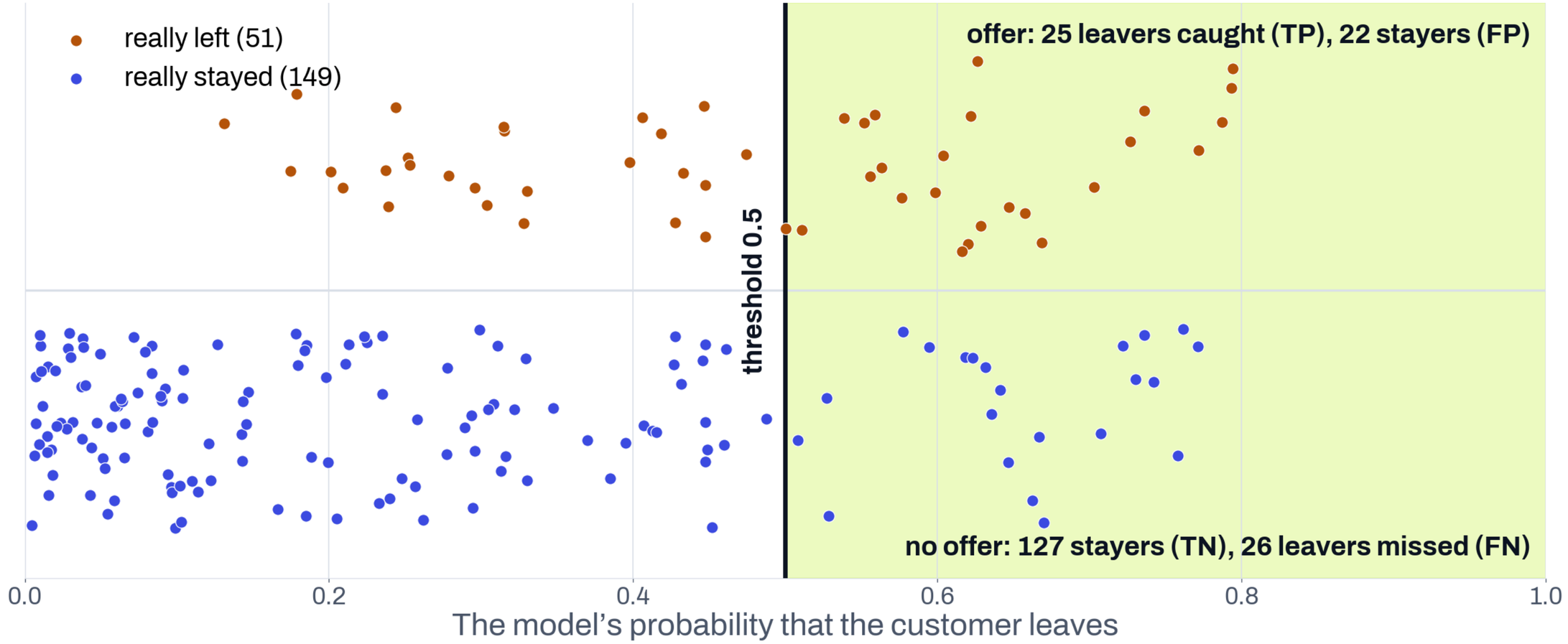
Suggested time · 6 minutes

THE RULE

Logistic regression gives a probability. 0.5 is BigQuery ML's default cut. The costs of each mistake should set yours.

200 HELD-OUT CUSTOMERS · EIGHT-FEATURE MODEL · THRESHOLD 0.5

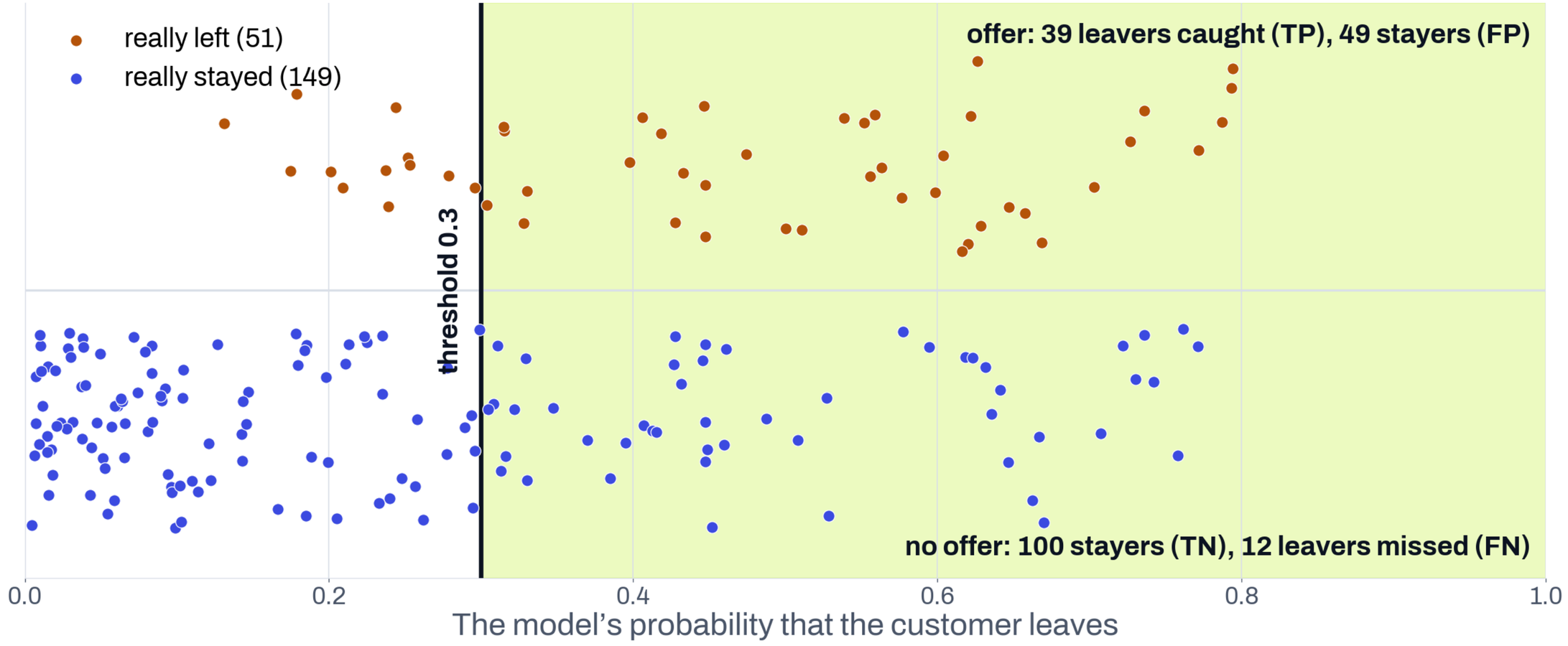
Right of the line, an offer



Each dot is a customer the model did not train on. Leavers above the axis, stayers below.

200 HELD-OUT CUSTOMERS · THRESHOLD 0.3

Move the line: more offers



The same 200 probabilities. Only the threshold changed.

INSTRUCTOR DO

Step 3.4

The confusion matrix, one row at a time

Four counts, in the retention team’s words.

Suggested time · 10 minutes

DEFINITION

A confusion matrix counts every row we score by its actual class and its predicted class: four cells for a yes / no target.

Positive class here: churned, the customers we act on. BigQuery puts actual classes in rows.

ONE HELD-OUT 20% · 1,407 CUSTOMERS · THRESHOLD 0.5 · 1 OF 2

Who stayed

	MODEL SAYS: STAYS	MODEL SAYS: LEAVES
REALLY STAYED	**TN** stayed · no offer **925**	**FP** stayed · offer wasted **109**
REALLY LEFT	**FN** left · no offer **178**	**TP** left · offer made **195**

The second word is what the model said; the first says whether it was right.

ONE HELD-OUT 20% · 1,407 CUSTOMERS · THRESHOLD 0.5 · 2 OF 2

Who left

	MODEL SAYS: STAYS	MODEL SAYS: LEAVES
REALLY STAYED	TN — stayed · no offer — 925	FP — stayed · offer wasted — 109
REALLY LEFT	FN — left · no offer — 178	TP — left · offer made — 195

Accuracy, the share it got right: (925 + 195) ÷ 1,407 = 0.80.

EVERYONE DO

Step 3.5

Read the matrix

Six questions from four counts.

Suggested time · 8 minutes

THRESHOLD 0.5 · TN 925 · FP 109 · FN 178 · TP 195

Six questions, four counts

#	Question
1	How many offers did the model make?
2	How many offers went to customers who would have stayed?
3	How many leavers got no offer?
4	What share of all customers did it get right?
5	Of those who got an offer, what share would have left?
6	Of those who left, what share got an offer?

THE ANSWERS

Six answers

#	Answer
1	195 + 109 = 304
2	109 (FP)
3	178 (FN)
4	(925 + 195) ÷ 1,407 = 0.80: accuracy
5	**195 ÷ 304 = 0.64: precision**
6	**195 ÷ 373 = 0.52: recall**

INSTRUCTOR DO

Step 3.6

Precision and recall

Two questions the retention team asks.

Suggested time · 7 minutes

DEFINITION

Precision is TP / (TP + FP): of the customers flagged, the share who really churned.

"Of the customers we make an offer to, how many would have left?" Down the "says Yes" column: 195 ÷ 304 = 0.64.

DEFINITION

Recall is TP / (TP + FN): of the customers who churned, the share the model flagged.

"Of the customers we lose, how many did we reach?" At 0.5: 195 ÷ 373 = 0.52.

PART 03

Many features, judged honestly

Eight features, the majority guess, the matrix in SQL, ROC and PR AUC, and the threshold.

EVERYONE DO

Step 4.1

Train with eight features

Same split, more to learn from.

Suggested time · 10 minutes

labs/D12/02-Evr-Many-Features/README.md

BLOCK 1 · TRAIN

Eight features, 20% held aside

Contract, InternetService, PaymentMethod: text, one-hot encoded for you.

```
CREATE OR REPLACE MODEL ml.churn_model
OPTIONS (
  model_type = 'LOGISTIC_REG',
  input_label_cols = ['Churn'],
  data_split_method = 'RANDOM',
  data_split_eval_fraction = 0.2
) AS
SELECT tenure, MonthlyCharges, Contract, InternetService,
       PaymentMethod, SeniorCitizen, PaperlessBilling, TechSupport,
       Churn
FROM ml.telco;
```

EVERYONE DO

Step 4.2

Score it against the majority guess

The scores BigQuery prints, and a guess that uses no features.

Suggested time · 14 minutes

labs/D12/02-Evr-Many-Features/README.md

Five scores in one row

ML.EVALUATE: accuracy, precision, recall, F1 and ROC AUC at threshold 0.5.

```
SELECT ROUND(accuracy, 3) AS accuracy, ROUND(precision, 3) AS precision,
       ROUND(recall, 3) AS recall, ROUND(f1_score, 3) AS f1,
       ROUND(roc_auc, 3) AS roc_auc
FROM ML.EVALUATE(MODEL ml.churn_model);
```

→ accuracy ≈ 0.80 · precision ≈ 0.64 · recall ≈ 0.52

ROC AUC ≈ 0.84. Each random split moves these a little.

Say "No" for everyone

COUNTIF: the share who stayed is the accuracy of always saying No.

```
SELECT ROUND(COUNTIF(Churn = 'No') / COUNT(*), 3)
       AS majority_accuracy
FROM ml.telco;
```

→ 1 row · 0.735

THE MODEL AGAINST THE MAJORITY GUESS

Accuracy hides the leavers

	Accuracy	Leavers found (recall)
Always say No	0.735	0
The model, threshold 0.5	≈ 0.80	**≈ 0.52**

When one class is rare, accuracy can look good while the model misses what matters.

Four counts, from BigQuery

ML.CONFUSION_MATRIX: the held-aside rows, at threshold 0.5.

```
SELECT *
FROM ML.CONFUSION_MATRIX(MODEL ml.churn_model);
```

```
→ 2 rows · expected_label No / Yes · columns No, Yes
```

INSTRUCTOR DO

Step 4.3

ROC AUC and PR AUC: how well it ranks

One number for every threshold at once, asked two ways.

Suggested time · 9 minutes

DEFINITION

The ROC curve plots, for every threshold, the share of churners flagged (TPR, recall) against the share of stayers wrongly flagged (FPR).

ROC AUC is the area under it: the chance a random churner scores higher than a random stayer.

Every threshold, in rows

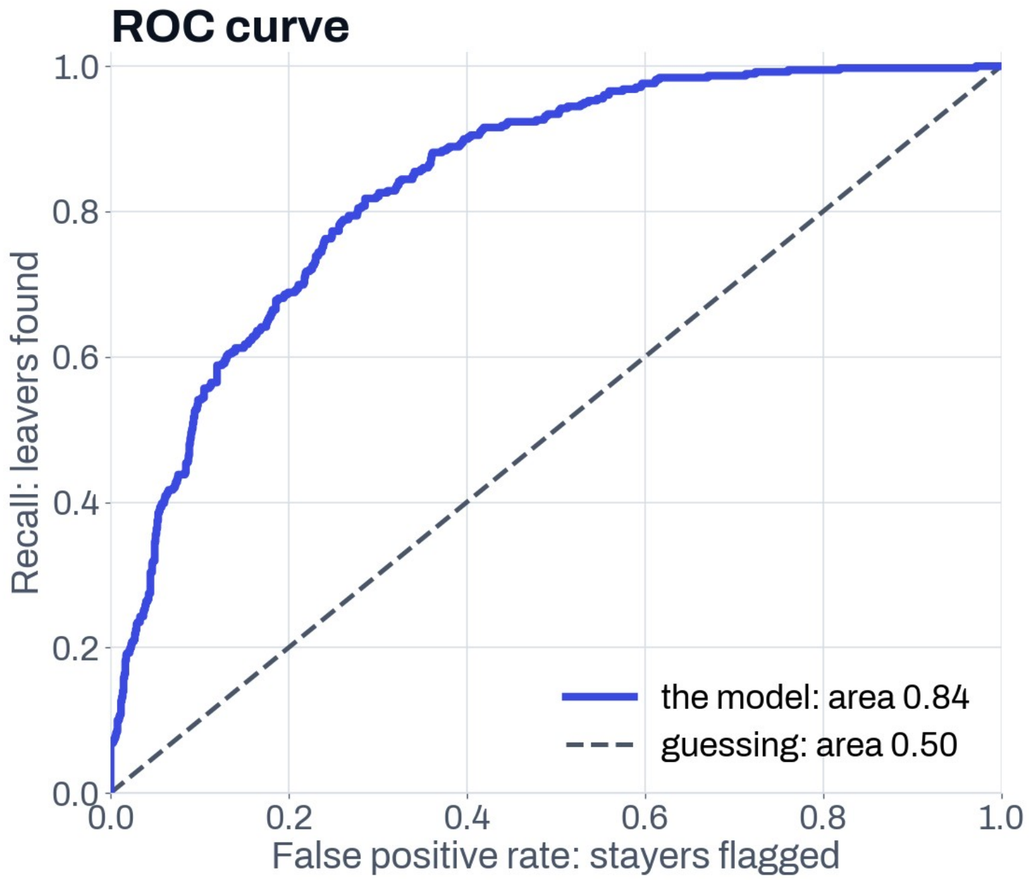
ML.ROC_CURVE: recall and false positive rate per threshold.

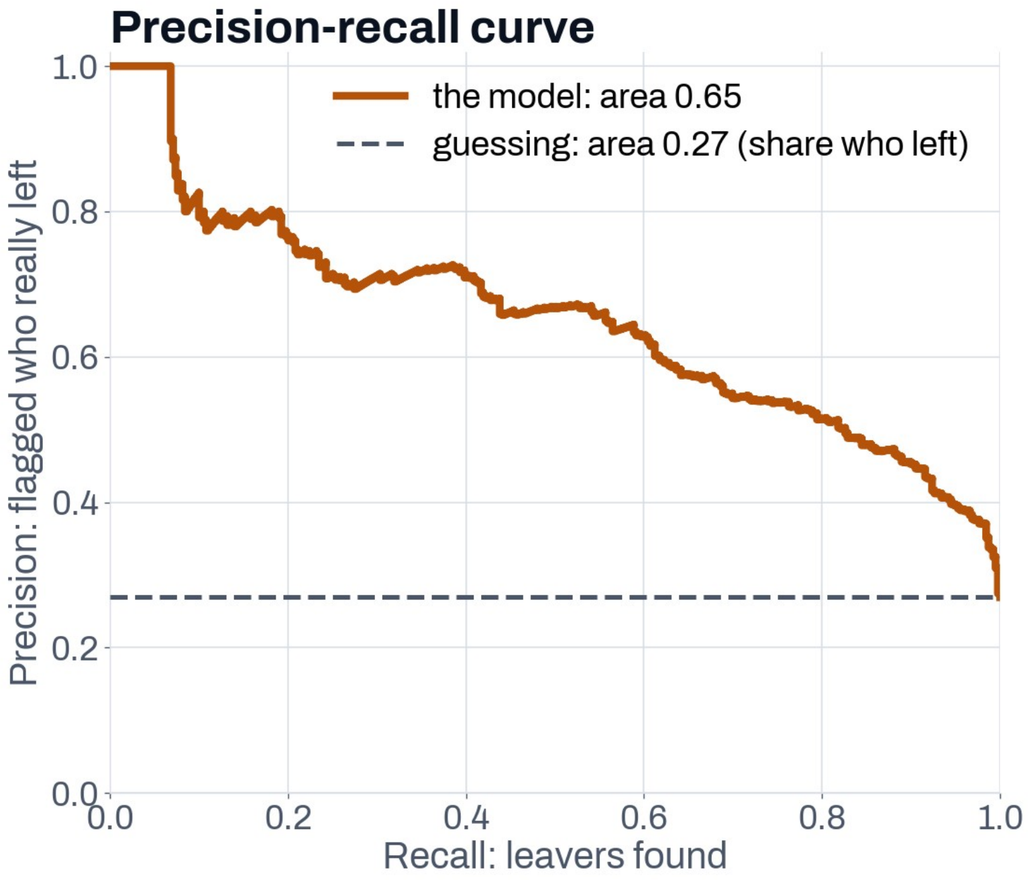
```
SELECT ROUND(threshold, 2) AS threshold, ROUND(recall, 3) AS recall,
       ROUND(false_positive_rate, 3) AS false_positive_rate
FROM ML.ROC_CURVE(MODEL ml.churn_model)
ORDER BY threshold;
```

→ lower threshold: more recall, more false alarms

ONE HELD-OUT 20% · EIGHT-FEATURE MODEL

Two curves, two questions





Dashed lines: what guessing scores. For ROC always 0.5; for precision-recall, the share who left.

TWO AREAS UNDER TWO CURVES

ROC AUC and PR AUC

	ROC AUC	PR AUC
Question	does a leaver get a higher probability than a stayer?	of the customers we flag, how many really leave, as we find more leavers?
Guessing scores	0.5	**the share who left: 0.265**
Our model	≈ 0.84	≈ 0.64
In BigQuery	roc_auc in ML.EVALUATE	from ML.ROC_CURVE (lab 03)

Compare each area with its own guessing line, never with each other.

EVERYONE DO

Step 4.4

Move the threshold

The same model, scored at 0.3, 0.5 and 0.7.

Suggested time · 12 minutes

labs/D12/02-Evr-Many-Features/README.md

One option: threshold

`STRUCT(0.3 AS threshold)`: a setting passed to an ML function.

```
SELECT '0.3' AS threshold, ROUND(precision, 2) AS precision, ROUND(recall, 2) AS recall
FROM ML.EVALUATE(MODEL ml.churn_model, STRUCT(0.3 AS threshold))
UNION ALL
SELECT '0.5', ROUND(precision, 2), ROUND(recall, 2)
FROM ML.EVALUATE(MODEL ml.churn_model, STRUCT(0.5 AS threshold))
UNION ALL
SELECT '0.7', ROUND(precision, 2), ROUND(recall, 2)
FROM ML.EVALUATE(MODEL ml.churn_model, STRUCT(0.7 AS threshold));
```

ONE HELD-OUT 20% · 1,407 CUSTOMERS

Precision against recall

Threshold	Offers	Leavers caught	Precision	Recall
0.3	≈ 538	≈ 286	≈ 0.53	≈ 0.77
0.5	≈ 304	≈ 195	≈ 0.64	≈ 0.52
0.7	≈ 93	≈ 72	≈ 0.78	≈ 0.19

Your counts differ a little: each random 20% is a different 1,400 customers.

EVERYONE DO

Step 4.5

Who does it miss?

The leavers the model said would stay.

Suggested time · 10 minutes

labs/D12/02-Evr-Many-Features/README.md

Left, but the model said No

predicted_Churn = 'No' among customers who left.

```
SELECT Contract, COUNT(*) AS missed_leavers,
       ROUND(AVG(tenure), 1) AS avg_tenure
FROM ML.PREDICT(MODEL ml.churn_model, (SELECT * FROM ml.telco))
WHERE Churn = 'Yes' AND predicted_Churn = 'No'
GROUP BY Contract;
```

→ the misses: longer tenure, yearly contracts

REVIEW

Time's up. Let's review.

What the numbers say

Solved: labs/D12/02-Evr-Many-Features/Solved/many_features.sql

THE ANSWER

Better than the guess, and the threshold is still open

Question	Answer
Better than saying No for everyone?	yes: it finds about half the leavers; the guess finds none
How well does it rank?	ROC AUC ≈ 0.84 (tenure alone ≈ 0.74)
Which threshold?	not decided yet: it depends on what each mistake costs

Break

30 minutes

After the break: what the model learned, what each mistake costs, and which customers to call.

PART 04

Read it, cost it, decide

What it learned, what mistakes cost, a rare Yes, other models, and a decision.

EVERYONE DO

Step 5.1

Read what it learned

Signs first: which way does each feature push?

Suggested time · 10 minutes

labs/D12/02-Evr-Many-Features/README.md

BLOCK 8 · ML.WEIGHTS

Which way does each feature push?

weight above 0 pushes toward Yes (leaves); below 0 toward No (stays).

```
SELECT processed_input, ROUND(weight, 3) AS weight, category_weights
FROM ML.WEIGHTS(MODEL ml.churn_model);
```

WHAT THE WEIGHTS SAY

Toward leaving, and toward staying

Pushes toward leaving	Pushes toward staying
month-to-month contract	one- or two-year contract
fiber optic internet	longer tenure
paying by electronic check	having tech support
paperless billing; senior citizen	no internet service

The same customer, two contracts

UNION ALL builds two customers the model has never seen.

```
SELECT Contract, ROUND(p.prob, 2) AS prob_leave
FROM ML.PREDICT(MODEL ml.churn_model,
  (SELECT 12 AS tenure, 70.0 AS MonthlyCharges, 'Month-to-month' AS Contract,
     'Fiber optic' AS InternetService, 'Electronic check' AS PaymentMethod,
     0 AS SeniorCitizen, 'Yes' AS PaperlessBilling, 'No' AS TechSupport
   UNION ALL SELECT 12, 70.0, 'Two year', 'Fiber optic', 'Electronic check', 0, 'Yes', 'No')),
  UNNEST(predicted_Churn_probs) AS p WHERE p.label = 'Yes';
```

→ month-to-month ≈ 0.65 · two year ≈ 0.31

INSTRUCTOR DO

Step 5.2

What does each mistake cost?

Invented costs, said to be invented: the threshold follows from them.

Suggested time · 10 minutes

INVENTED COSTS · AN OFFER 50 · A LOST CUSTOMER 500 · 1 IN 3 KEPT

Price each cell of the matrix

Cell	What happens	Value
TP: left, got an offer	1 in 3 stays: 500 ÷ 3 saved, minus 50	≈ +117
FP: stayed, got an offer	the offer was wasted	−50
FN: left, no offer	lost, and we never tried	0 (the loss happens either way)
TN: stayed, no offer	nothing happens	0

THE SAME THREE THRESHOLDS, PRICED

The costs choose the threshold

Threshold	Offers	Leavers caught	Value of the campaign
0.3	≈ 538	≈ 286	**≈ +20,800**
0.5	≈ 304	≈ 195	≈ +17,300
0.7	≈ 93	≈ 72	≈ +7,400

Value = leavers caught × 500 ÷ 3 – offers × 50. Change the costs and the best threshold moves.

INSTRUCTOR DO

Step 5.3

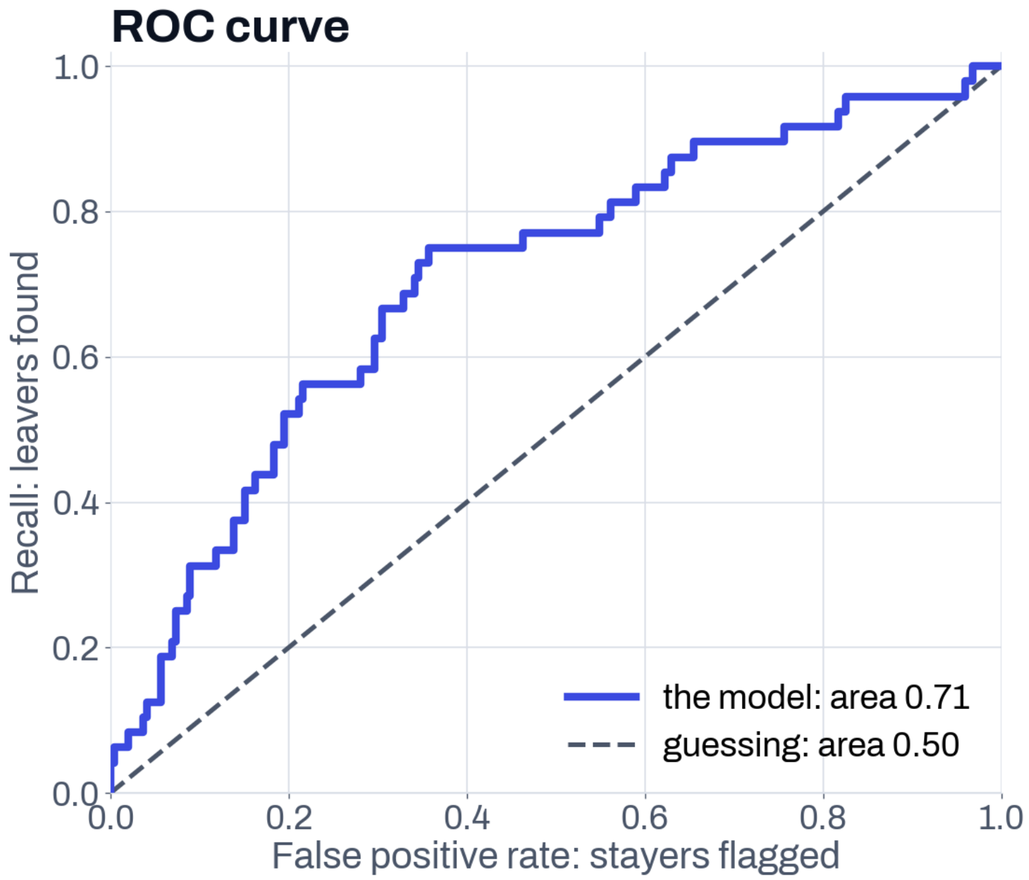
When the Yes is rare: metrics and class weights

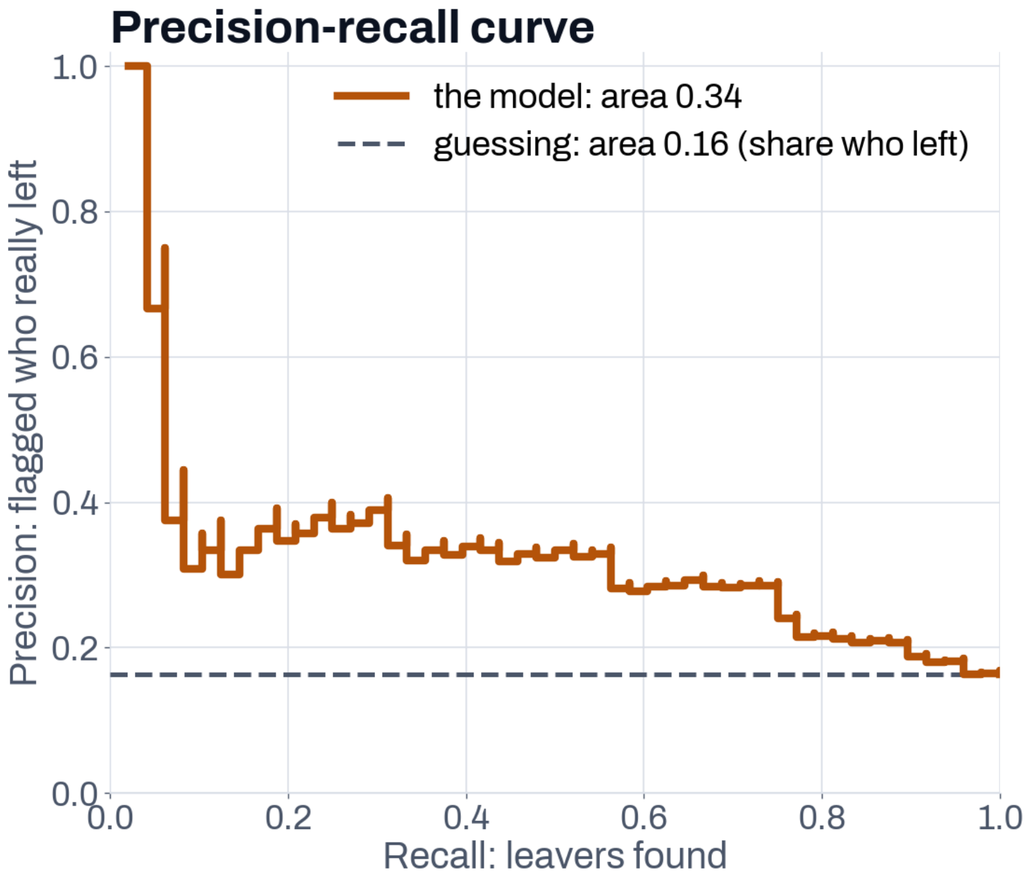
The most common problem in real data; and the CREATE MODEL page again.

Suggested time · 8 minutes

LAB 04 DATA · 1,470 EMPLOYEES · 16% LEAVE · ONE HELD-OUT 20%

A rare Yes: PR shows what you will get





Same model, same 294 employees. Guessing scores 0.5 on the left, 0.16 on the right.

WHICH NUMBER, WHEN THE YES IS RARE

Metrics for a rare Yes

Metric	What it tells you	In BigQuery
Accuracy	misleading: saying No to everyone scores high	ML.EVALUATE
Precision, recall	flagged who leave; leavers found	ML.EVALUATE
F1	precision and recall in one number, at one threshold	f1_score
ROC AUC	ranking; guessing scores 0.5 whatever the share	roc_auc
PR AUC	ranking judged on the Yes; guessing scores the share	from ML.ROC_CURVE
You will also meet	balanced accuracy, MCC, F-beta, precision in the top k	in Python

DEFINITION

AUTO_CLASS_WEIGHTS = TRUE: makes each rare Yes count as much, in training, as the many Nos together.

LAB 04 · EMPLOYEES · THRESHOLD 0.5 · TYPICAL VALUES

What class weights change, and what they do not

Model	Recall	F1	ROC AUC	PR AUC
plain	≈ 0.02	≈ 0.04	≈ 0.69	≈ 0.32
AUTO_CLASS_WEIGHTS	**≈ 0.67**	**≈ 0.36**	≈ 0.69	≈ 0.31

Guessing scores 0.16 on PR AUC here. A lower threshold on the plain model does much the same as the weights.

CREATE MODEL · LOGISTIC REGRESSION OPTIONS

The same page, two new options

Options	What they change	Lab
AUTO_CLASS_WEIGHTS, CLASS_WEIGHTS	how much a rare class counts in training	D12 04
threshold (in ML.EVALUATE, ML.PREDICT)	where a probability becomes Yes	D12 03
DATA_SPLIT_*, L1_REG, L2_REG, CATEGORY_ENCODING_METHOD …	as yesterday	D11 05–07
more than two classes in the label	multiclass logistic regression	D12 05

INSTRUCTOR DO

Step 5.4

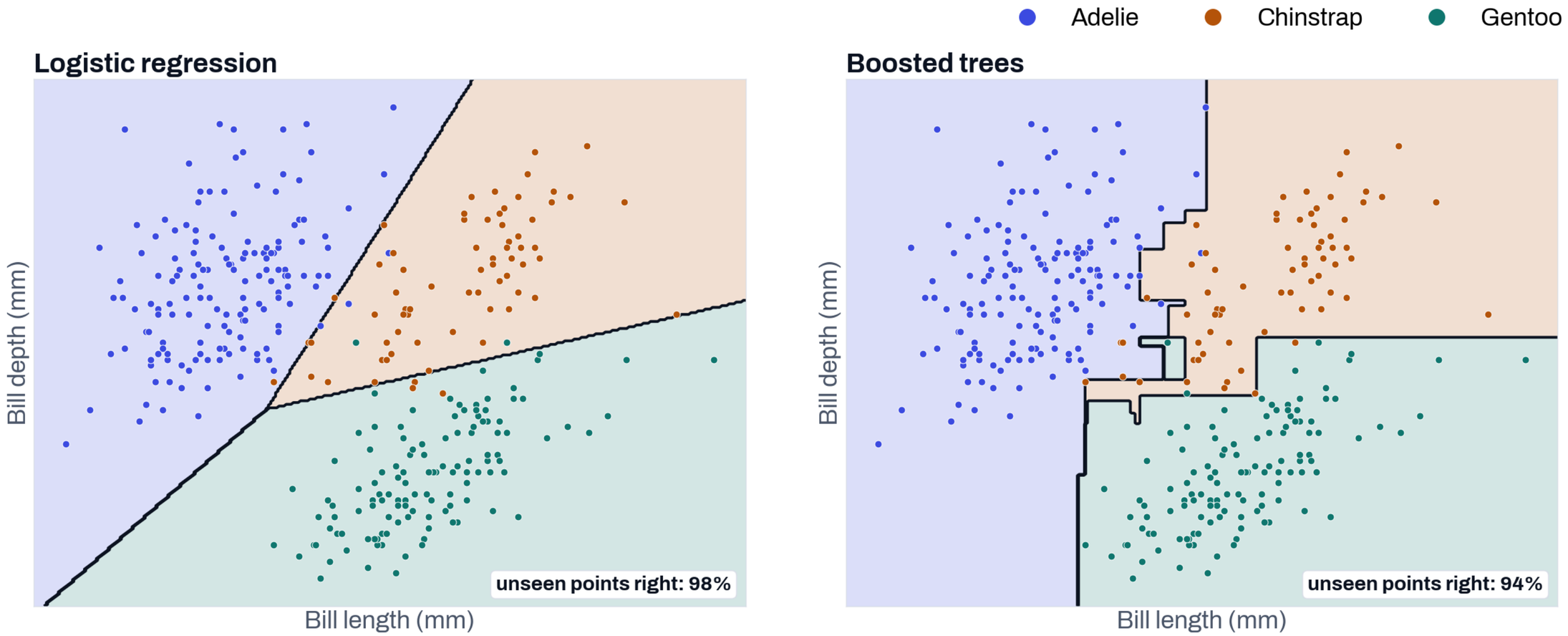
Beyond logistic regression

Straight boundaries and curved ones; what else BigQuery ML offers.

Suggested time · 7 minutes

PALMER PENGUINS · THREE SPECIES · BILL LENGTH AND DEPTH

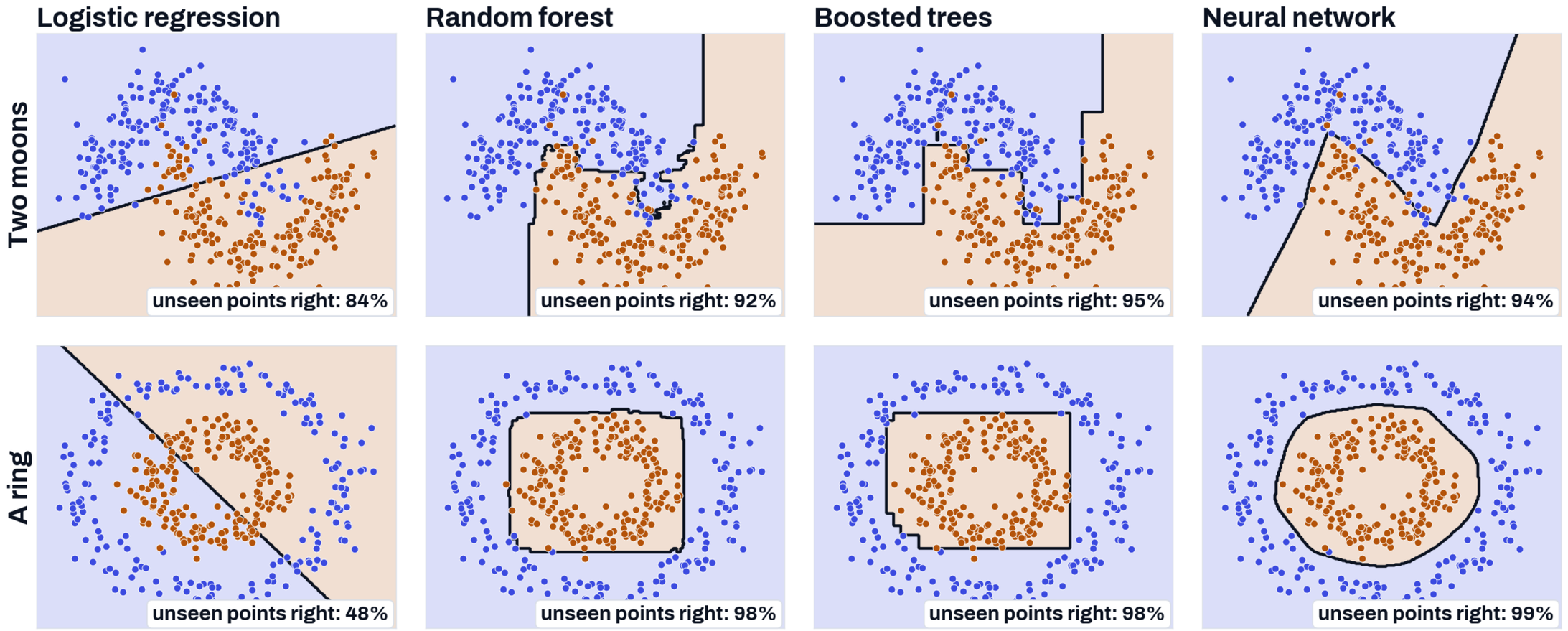
When straight lines are enough



Each panel trained on 70% of the penguins, scored on the other 30%.

TWO MADE-UP SHAPES · 400 POINTS EACH

When the boundary curves



Trees build boundaries from steps; a neural network bends them. Scored on 30% held aside.

CREATE MODEL · CLASSIFICATION

What BigQuery ML offers for classification

Model	model_type	Reach for it when
Logistic regression	**LOGISTIC_REG**	a fast, readable baseline: start here
Boosted trees (XGBoost)	BOOSTED_TREE_CLASSIFIER	tables with curved boundaries; often the strongest
Random forest	RANDOM_FOREST_CLASSIFIER	tables; many trees averaged, little tuning
Deep neural network	DNN_CLASSIFIER	complex patterns in large tables
Wide & deep	DNN_LINEAR_COMBINED_CLASSIFIER	many sparse categories, as in recommendations
AutoML	AUTOML_CLASSIFIER	Google tries models and tuning for you
Gemini	AI.CLASSIFY (a function)	sorting free text into your categories, no training

Only LOGISTIC_REG trains inside BigQuery. The others run on Vertex AI and need billing.

Other classifiers you will meet

Classifier	How it decides
Decision tree	a few yes-or-no questions; easy to read, easy to overfit
k-nearest neighbours	the label most common among the closest examples
Naive Bayes	probabilities from counts; a classic for spam
Support vector machine	the widest gap between classes; curves with kernels
XGBoost, LightGBM	boosted trees in Python, the same idea as BOOSTED_TREE

PARTNER DO

Which customers would you call?

The retention team can make about 500 offers this month. Which threshold, and which customers?

Suggested time

10 minutes

`labs/D12/02-Evr-Many-Features/README.md`

PARTNER DO

Answer the retention team in three sentences

INSTRUCTIONS	They can make about 500 offers this month. Which threshold would you use, and why? Which kinds of customer will get most of the offers? What should they know about the leavers you will miss?
HINT	Use the threshold table and the weights.
BONUS	If an offer cost 100 instead of 50, which threshold would the costs choose?

Full steps and Check yourself: labs/D12/02-Evr-Many-Features/README.md

REVIEW

Time's up.
Let's review.

Which customers would you call?

`Solved: labs/D12/02-Evr-Many-Features/Solved/many_features.sql`

A SAMPLE ANSWER

Threshold 0.3: about 540 offers

Threshold 0.3 gives the most value: about 540 offers, reaching three in four leavers. Most go to new month-to-month customers on fiber. Long-tenure customers on yearly contracts are the misses.

STUDY LIST · IN CLASS AND IN THE LABS

BigQuery ML functions today

In class	What it gives you	Activity
ML.PREDICT	each row’s predicted class and its probabilities	01, 02
ML.WEIGHTS	the numbers the model learned, feature by feature	01, 02
ML.EVALUATE	accuracy, precision, recall, F1, log loss, ROC AUC	02
ML.CONFUSION_MATRIX	counts of right and wrong answers, by class	02
ML.ROC_CURVE	recall, false positive rate and counts per threshold	02

New in the labs	What it gives you	Lab
ML.METRICS	scores from actual and predicted columns (Preview)	03
ML.GLOBAL_EXPLAIN	how much each feature drives predictions, on average	05

EVERY MODULE 3 FUNCTION · resources/bigquery-ml-functions.md

CLOSE

What you can do now

- Train LOGISTIC_REG; read a probability
- Turn a probability into a decision
- Read a confusion matrix in business words
- Beat the majority guess; let costs pick

NEXT · D13 Grouping without a label. A1 due Wednesday.

LAB

AFTER CLASS · SELF-GUIDED

Classification, option by option

TASK

Three short labs: thresholds and curves; imbalance and class weights; three classes.

FILES

`labs/D12/03-Lab-Thresholds-And-Curves/README.md`

`labs/D12/04-Lab-Imbalance-And-Class-Weights/README.md`

`labs/D12/05-Lab-Three-Classes/README.md`

DONE WHEN

Each lab ends with a table of what you saw.

Optional self-study: lab 06, Nakheel churn end to end.